

Ryan Epperly

☎ (540)-818-0146 | ✉ ryan7820@gmail.com | 🔗 linkedin.com/in/ryanepperly | 🇺🇸 U.S. Citizen

EDUCATION

Virginia Tech

Jan 2026 – May 2028

Bachelor of Science in Aerospace Engineering

- **Specialization:** Dynamics, Control, and Estimation | **Minor:** Mathematics

New River Community College

Aug 2024 – Dec 2025

Associate of Science (A.S), Engineering | Magna Cum Laude

SKILLS & CERTIFICATIONS

Software & Programming : SolidWorks, AutoCAD, Autodesk Inventor, Ansys, MATLAB, Python, Java, GNU Octave, LaTeX, PrusaSlicer

Hardware & Testing : 3D Printing (Bambu Lab P2S, FDM/SLA), CNC Machining, Plasma, Laser & Waterjet Cutters, Glowforge, Mecmesin

Certifications : FAA UAS Safety Test, NASA Open Science 101, Ansys Topology Optimization, FAA NAFI SMS, FEMA (IS-66, IS-922.A), UCAR Satellite Remote Sensing

EXPERIENCE

Butler Parachute Systems

May 2026 – Aug 2026

Engineering Intern

- Designed a line continuity tool in SolidWorks, optimizing geometry to eliminate sharp edges and prevent cord snagging.
- Executed 150+ Mecmesin/hydraulic tensile tests and performed 10 operational drop tests with a custom load cell amplifier enclosure.
- Engineered a quality control program referencing PIA documents and MIL-STD-849D, improving inspection efficiency by 20%.
- Fabricated government-regulated hardware using a Glowforge, plasma, and laser cutters; rigged 2 personnel parachutes with an FAA master rigger.

GEOINT ISAC

May 2026 – Oct 2026

Space Systems Intern

- U.S. Space Force Project: Programmed a Python-based AI predictive model for the 460th CES to track and forecast critical infrastructure vulnerabilities based on age, exposure, and proximity.
- Engineered predictive analytics algorithms and automated data collection to define statistical thresholds for highly accurate weekly, monthly, and annual risk forecasts.
- Federal ESF Project: Evaluated 15 primary Federal Emergency Support Functions (ESFs), analyzing how geospatial intelligence guides disaster response by mapping infrastructure damage.
- NASA Joint Project: Executed meteorological risk analysis and geospatial mapping for Project Gestalt to advance global critical infrastructure resilience.

Blackwing Space

Mar 2026 – Aug 2026

Regulatory Engineering Intern

- Analyzed CubeSat systems and compiled FCC technical materials to ensure compliance and expedite satellite mission approvals.
- Verified physical component tolerances and 3D models (SolidWorks/Fusion 360) against strict aerospace licensing standards.

PROJECTS & RESEARCH

Starlink Sentinel: Autonomous LEO Collision Avoidance

May 2026

- Developed a 6-DOF orbital dynamics simulator in MATLAB using an RK4 numerical integrator to model autonomous collision avoidance.
- Implemented an Extended Kalman Filter (EKF) to process noisy ground radar measurements and predict incoming debris trajectories.

Comparative Analysis of Radiation Shielding Materials

Jan 2026 – May 2026

- Evaluated radiation shielding efficacy of projected Martian construction materials by developing MATLAB simulations and 3D Inventor models.
- Determined that polyethylene rapidly reduces radiation exposure to safe baseline levels at a minimal shield thickness of 7 cm.

Space Law and Policy (VASTS at NASA)

Oct 2023 – Jun 2024

- Drafted a legal and regulatory framework proposal for a crewed Mars mission with NASA engineers, outlining planetary protection protocols, propulsion safety requirements, and radiation limits.